Robin Hur

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Education

University of Virginia, School of Engineering and Applied Science **Charlottesville, VA**
B.S. in Mechanical Engineering, GPA: 3.75 **Expected May 2027**

- **Relevant Coursework:** Statics, Strength of Materials, Thermodynamics, Fluid Mechanics, Dynamics, Material Science, Computational Methods, Mechatronics

Skills

- **Design & Modeling:** SOLIDWORKS (CSWA), Fusion 360, GD&T, Technical Documentation, FEA
- **Analysis & Control:** Python, MATLAB, Simulink, CFD, Tecplot 360, Molecular Dynamics (AMBER), PID Control
- **Hardware & Tools:** Embedded Microcontrollers, Git, Linux, High-Performance Computing (HPC), SLURM
- **Languages:** English (native), Korean (fluent), French (intermediate)

Experience

Research Assistant, *UVA Dept. of Mechanical and Aerospace Engineering* **May 2026 – Present**

- Conducted CFD simulations and numerical optimization on bio-inspired underwater robotic systems to evaluate hydrodynamic performance and maximize propulsion efficiency.
- Executed parallel computing runs on university HPC clusters, processing large-scale 3D flow field datasets for hydrodynamic post-processing and thrust/drag analysis.
- Communicated research findings through weekly technical reports and design reviews to drive project milestones.

Research Assistant, *UVA Dept. of Materials Science and Engineering* **Feb 2026 – Present**

- Developed closed-loop feedback control in Python to dynamically regulate N_2 flow and relative humidity.
- Engineered a Python-driven closed-loop PID control system integrating Alicat mass flow controllers and Monarch probes, maintaining relative humidity within $\pm 2\%$ over 20+ hours of continuous testing.
- Advanced environmental cracking test setups for high-strength aerospace aluminum alloys supporting research sponsored by NASA, Lockheed Martin, and the F-35 program.

Research Assistant, *UVA Dept. of Chemical Engineering* **Aug 2024 – May 2026**

- Executed all-atom molecular dynamics (MD) simulations in AMBER to analyze cross-linking mechanisms and atomic-level structural evolution during hydrogel gelation.
- Developed automated Python data pipelines with Git version control to process simulation trajectories in VMD.
- Managed data architecture and storage pipelines for 10+ TB of simulation trajectories across 100+ HPC configurations.

Research Outputs:
Co-Author, *A Methodological Approach to Relate Pentapeptide Sequence, Atomistic Self-Assembly, and Gelation Behavior*, ACS Biomaterials Science & Engineering **Jun 2026**

Teaching Assistant, *UVA Dept. of Electrical and Computer Engineering* **Jan 2026 – May 2026**

- Supported undergraduate students during embedded systems laboratories and office hours, reinforcing microcontrollers, C programming, and circuit debugging concepts.
- Evaluated lab assignments and collaborated with course instructors to refine technical course materials.

Bridge Program Counselor, *UVA School of Engineering and Applied Science* **Jul – Aug 2025**

- Mentored 38 first-year engineering students as part of a 6-member team during a 3.5-week residential program, guiding academic development and tutoring pre-calculus.

Projects

Mechatronics and Robotics Society, *Mechanical Subteam* **Aug 2025 – Dec 2025**

- Designed CAD assemblies for a NASA Lunabotics competition robot focused on regolith excavation and material handling.
- Researched vibration-assisted granular flow to optimize excavation subsystem efficiency in simulated lunar soil.

Potential Energy Propulsion Racer, *Project Member* **Nov 2025**

- Designed and fabricated a 3D-printed, potential-energy-powered vehicle using CAD modeling and mechanical transmissions.
- Authored technical reports and curated complete inventory of CAD assemblies, parts, and bill of materials.